



Outline of the lecture

The history of Informatics & HCI

Our roots and evolution

Conceptual Foundations

Information Processing | Gestalt Theory | Functional Approaches

Action Research & Praxeology

Work(place) studies & ethnography | Contextual Approaches |
Praxeology & Appropriation | Activity Theory

Structuration & Infrastructuring

A holistic look at Human-System-Interaction

HCI in Practice

Designing | Prototyping | Evaluation

Affordances: Signifiers and Concept Model

Especially in the field of usability, “affordance” is an expression often used only semi-correctly. Change in meaning over time:

Affordance

Possible actions

possible actions that an object or system allows a user to perform - a door affords (or allows) pushing or pulling, and a button affords pressing - can be ambiguous without additional clues

Signifiers

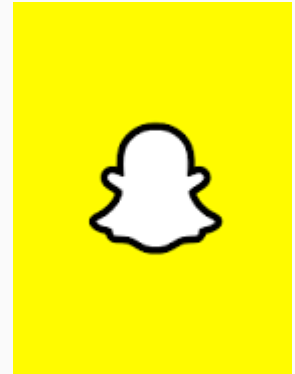
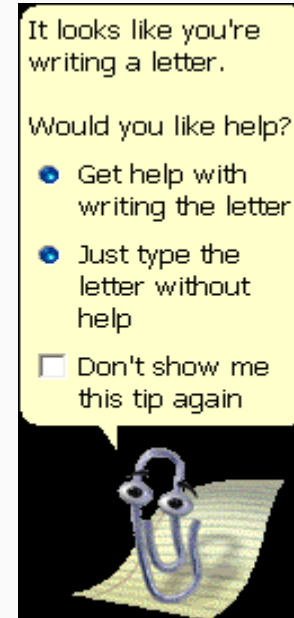
Cues indicating how to use an affordance

guide user's interaction by indicating where and how an action should be taken - a handle on door, an arrow may to swipe on a touchscreen - signifiers can be visual, textual, or even tactile - indicating the existence of affordance or intended use

Concept Model

Understanding: about how a system works

the design elements align well with the user's conceptual model, it makes the interaction intuitive - the design must supports the user in forming a conceptual model that matches the intended functionality



Affordances: Expression often used unclearly

Especially in the field of usability, “affordance” is an expression often used only semi-correctly. Change in meaning over time:

J.J. Gibson
1978

All possible actions

The Ecological Approach to Visual Perception: Co-evolution between creatures and environment ?
Affordance = **every** possible action that a specific creature has in relation to the observed object.

Don Norman
1988

“Suggestions” for actions

Affordance as the “**typical**” possible action for an object, e.g. “sitting” for a chair.
1999: Norman’s self correction: “Perceived Affordance” (describes his meaning in a far better way).

Bill Gaver
1991

Differentiated definition

Different categories of affordances, especially with the aim of dissolving the confusion caused by Gibson’s <-> Norman’s different understandings).



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“... the perceived or actual properties of the thing, primarily those fundamental properties that determine just how the thing could possibly be used... A chair affords (‘is for’) support and therefore affords sitting. A chair can also be carried. Glass is for seeing through, and for breaking.” (Norman, 1988)

“Affordances provide strong clues to the operations of things. Plates are for pushing. Knobs are for turning. Slots are for inserting things into. Balls are for throwing or bouncing. When affordances are taken advantage of, the user knows what to do just by looking: no picture, label, or instruction needed.” (Norman, 1988)



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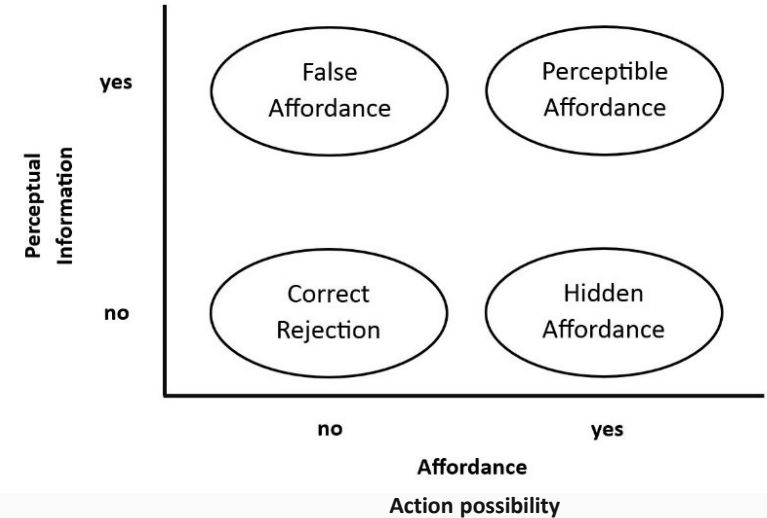
Different categories of affordances, especially with the aim of dissolving the confusion caused by Gibson's <-> Norman's different understandings). Examples:

Perceptible A.: Door perceived to be such

False A.: Wall painted (too) realistically to look like a door

Correct Rejection: Wall that doesn't look like a door

Hidden A.: Door behind wallpaper



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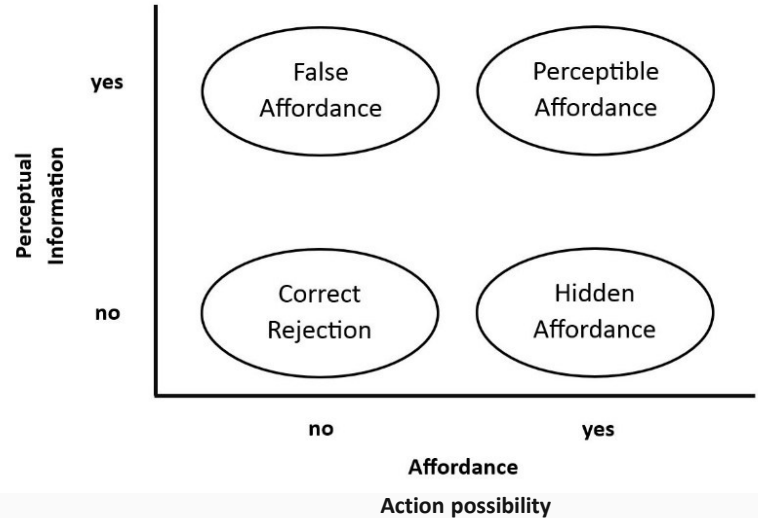
Correct Rejection: Wall that doesn't look like a door

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For AI
applications

Anticipatory affordance: predictive, personalization

Control affordance: user autonomy, agency



Affordances & Semiotics

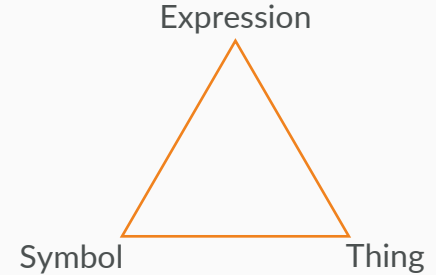
The perception of affordances is (also) linked to semiotics

Reality, perception and communication

We perceive real objects (things), then make ourselves a mental image (symbol) of these things and create words (expression) which we then use to speak to others about the image. There can easily be discrepancies between reality, imagination and description.



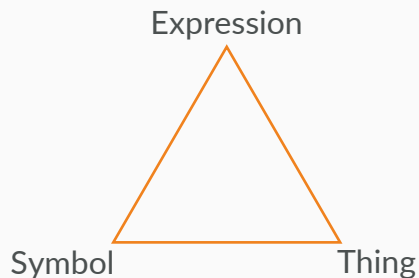
Examples of symbols:
Everywhere in this lecture...



Semiotic Triangle

Semiotics & Knowledge

How do we know what we are able to do?



Semiotic Triangle

How does the relation between expression, symbol and thing come about?

Know-how regarding the use of artefacts arises through the interplay of current reception and historical experience.

1

Sensory and motor functions

Senses: Seeing, hearing, smelling, tasting, haptics, movement; Actuators: Movement by muscle power

2

Interaction and communication

Questioning things through trial and error.
Development of use by other users.

3

Culture

Conventional patterns of use; similar forms; self-evidence and reliability of use

Cognitive instruments: Metaphors

Humans think and express themselves in meaningful analogies



Metaphors
everywhere...

In HCI, metaphors are often used to demonstrate **functionality**. The basis for this: the request for and use of metaphors in the communication with users, comparison of diverse metaphors for the analysis of difficulties in understanding and conflicts.



Finding suitable metaphors

The aim is often to demonstrate 'new' functionality without having an equivalent in the real world. However, the understanding of metaphors is shaped by the cultural and social background of designers and users.



Types of metaphor

Verbal / auditive / visual / compound / ...
Interface metaphors should combine what is already familiar with new functionalities



Interesting:

The dominance of navigational structures vs. The dominance of metaphors as semiotic signs

Cognitive instruments: Metaphors

Humans think and express themselves in meaningful analogies

Dominance of navigational structures

Navigational structures typically involve the use of clear and structured pathways or systems to guide or lead users through information, interfaces, or environments.

In website design, navigational structures are common. Websites often have clear menus and sitemaps that guide users to different sections and pages. For instance, a website might have a menu at the top with categories like "Home," "About Us," "Products," and "Contact." Users can navigate through these structured links to find the information they need.

Dominance of metaphors as semiotic signs

"metaphors" are being used as a way to represent and convey meaning. Metaphors involve comparing one thing to another to draw similarities and make a point. "Semiotic signs" refer to signs or symbols that carry meaning in a broader sense.

Smartphones use metaphors as semiotic signs extensively. Instead of spelling out "Phone," "Messages," or "Settings" in text, smartphones use icons like a telephone handset, speech bubble, or gear wheel. These icons serve as metaphors representing the functions they stand for. Users understand that the speech bubble icon is for messages and the gear wheel icon is for settings.

Examples for Metaphors



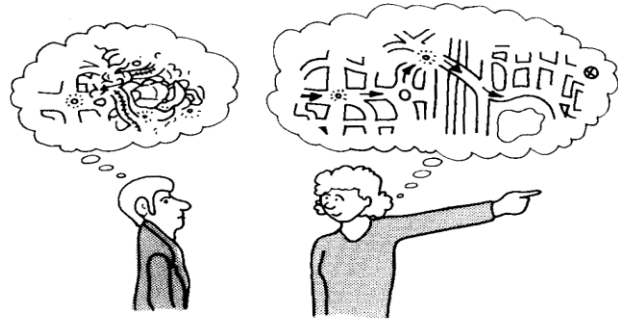
Drawing by Koren; © 1979 The New Yorker Magazine, Inc.

Table 7.1 Examples of applications and associated metaphors.

<i>Application area</i>	<i>Metaphor</i>	<i>Familiar knowledge</i>
Operating environment	The desktop	Office tasks, file management
Spreadsheets	Ledger sheet	Columnar tables
Object-oriented environments	Physical world	Real-world behaviour
Hypertext	Notecards	Flexible organization of structured text
Learning environments	Travel	Tours, guides, navigation
File storage	Piles	Categorizing objects in terms of urgency, projects and so on
Multimedia environments	Rooms (each associated with a different medium/task)	Spatial structure of buildings
Computer supported cooperative work	Multi-agents	Travel agents, butlers and other serving roles

Examples for Metaphors

<i>Application Area</i>	<i>Metaphor</i>
<i>ATM</i>	<i>Money Vending Machine</i>
<i>Augmented Reality</i>	<i>Magic Window</i>
<i>Virtual Reality</i>	<i>Digital Wonderland</i>
<i>Chatbot</i>	<i>Digital Assistant</i>
<i>Smart home</i>	<i>Nervous System of a House</i>
<i>Holograms</i>	<i>Glimpses from a Parallel Universe</i>
<i>Smart TV</i>	<i>Entertainment Hub</i>
<i>Smart Production</i>	<i>The Factory of the Future</i>



'If the organism carries a "small-scale model" of external reality and of its own possible actions within its head, it is able to try out various alternatives, conclude which is the best of them, react to future situations before they arise, utilise the knowledge of past events in dealing with the present and future, and in every way to react in a much fuller, safer, and more competent manner to emergencies which face it.'

(Fraik, 1943, p. 57)

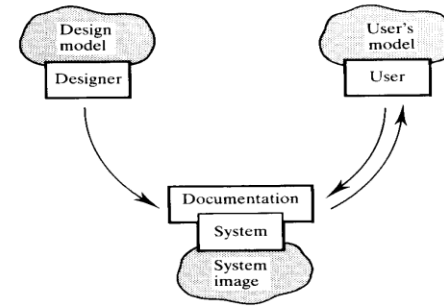


Figure 7.3 The design model, the user's model and the system image (Norman and Draper, 1986, p. 46).

Cognitive instruments: Mental Models

= Representation of reality and one's own options for action which can inform real or imaginary actions

HCI designers aim to design interfaces and interactions that align with users' mental models. This means making the system's behavior and structure intuitive and in line with how users expect it to work. By doing so, designers can reduce cognitive load and improve user satisfaction.

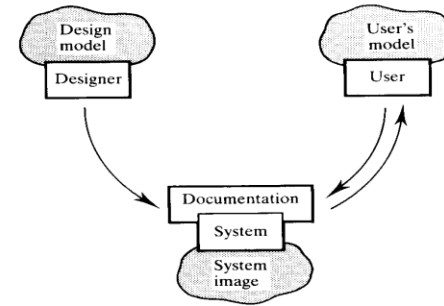


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Distributed Cognition

Knowledge at that time: Purely cognitive models probably won't be sufficient

"Cognition in the wild" (Hutchins 1995, 1990): Cognitive approaches only stem from individuals. Good: Human-computer interaction. Bad: Human-human interaction mediated by computers. Alternative: Distributed cognition as cognitive processing in the group.

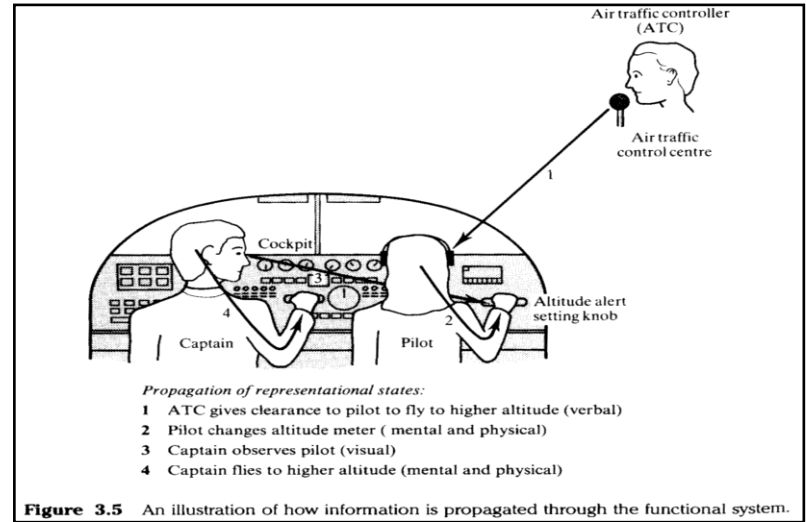


Focus of the analysis

Interplay of the various functional components and actors. A particular focus on the flow and transformation of information as well as on the inclusion of technical, social and organizational aspects. Additionally: the compilation of representational states of objects and media.



Example: Co-operation in the cockpit



Gestalt laws of grouping

Psychological basics

Gestalt laws

Habit / Expectation determines what we see.

how people form perceptions and give meaning

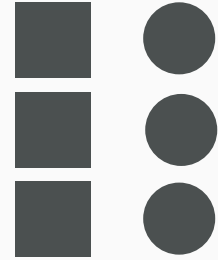
Gestalt-theory from 1920s cognition research: rules of experience regarding shape / colour / etc.

Over 100 “laws” with the capability of mutual influence (e.g. intensification by combination and /or multiple occurrence)

In HCI: Produce / avoid contexts of meaning, improve perception, reduce reaction time, improve search / recognition,...



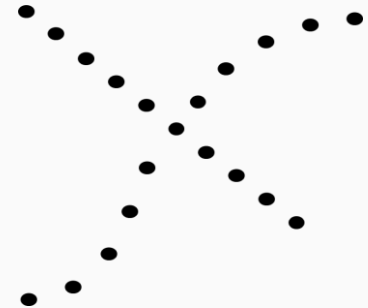
Law of Proximity



Law of Similarity



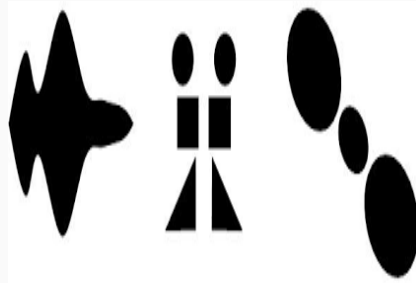
Law of Closure



Law of Continuity

Gestalt laws of grouping

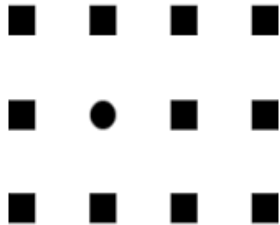
Psychological basics



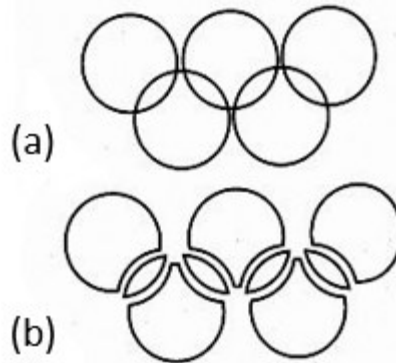
Law of Symmetry



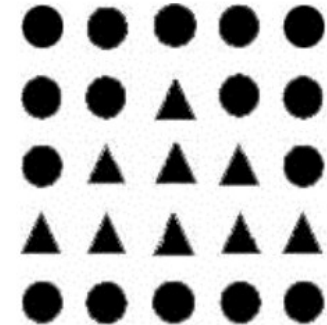
Law of Figure-Ground



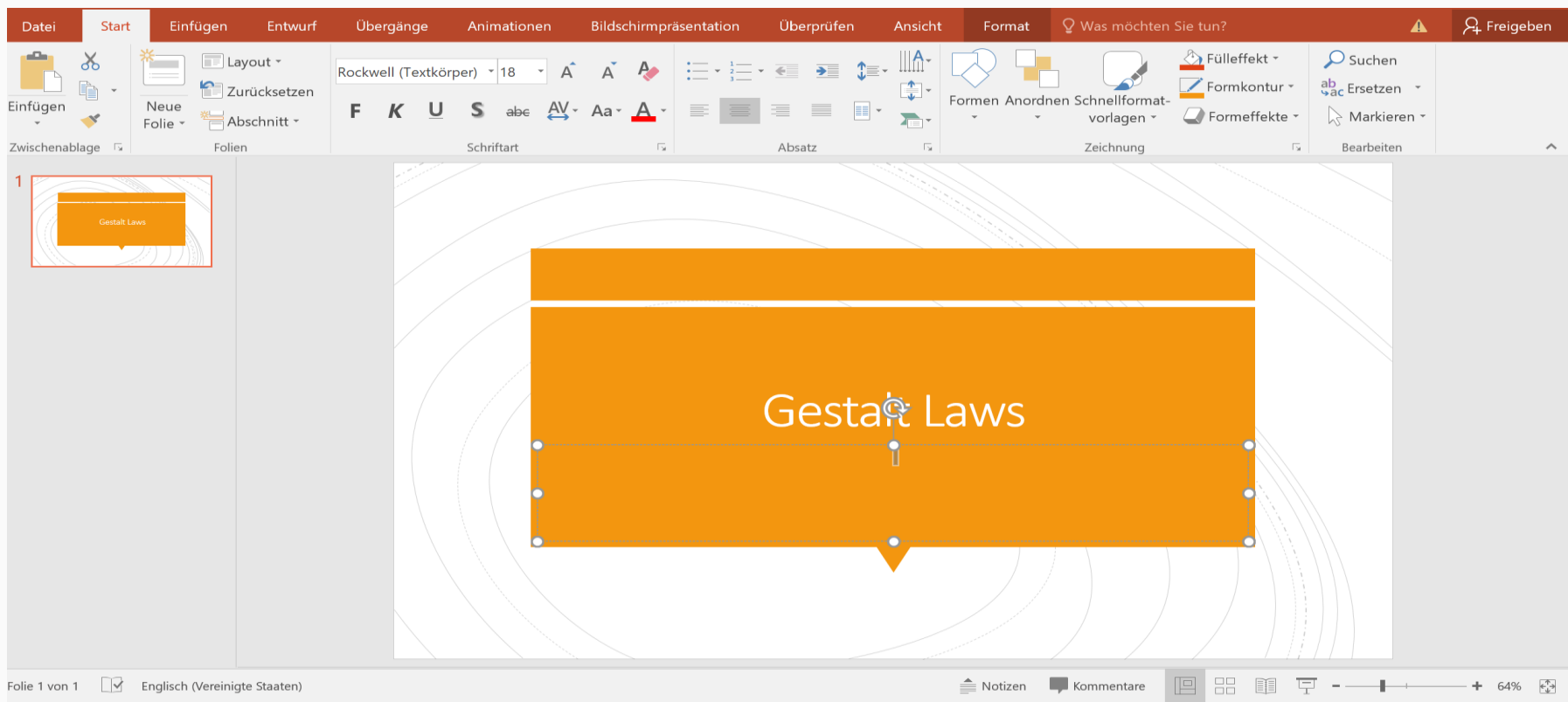
Law of Focal Point



Law of Prägnanz



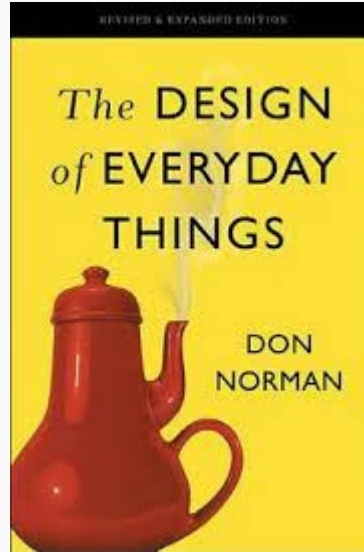
Law of Simplicity



Gestalt Laws?

Myriad examples can be found in (almost) every UI

Design must support functionality



<http://www.baddesigns.com/>

Interesting: <https://www.art.yale.edu/>